

Table 1: Pass-sum execution time (s, median per iteration; sum of pass medians, not full executable wall time) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with @BENCH adt_fields; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means the denominator is faster; **bold** marks >1.1×.

Program	ADT fields	SoA bufs	End-to-end					Fold passes					Map passes				
			Am (s)	Ai (s)	Sm (s)	Am/Sm	Ai/Sm	Am (s)	Ai (s)	Sm (s)	Am/Sm	Ai/Sm	Am (s)	Ai (s)	Sm (s)	Am/Sm	Ai/Sm
OctTree	16	9	0.8456	–	0.7832	1.08×	–	0.4816	–	0.3630	1.33×	–	0.3640	–	0.4202	0.87×	–
Compiler	9	8	0.3967	1.599	0.2670	1.49×	5.99×	0.1975	1.073	0.0526	3.76×	20.41×	0.1992	0.5255	0.2144	0.93×	2.45×
DBQuery	15	13	0.3218	0.3993	0.2865	1.12×	1.39×	0.1568	0.2196	0.0936	1.68×	2.35×	0.1650	0.1797	0.1929	0.86×	0.93×
DecisionTree	7	6	5.434	–	3.200	1.70×	–	5.434	–	3.200	1.70×	–	–	–	–	–	–
DomTree	15	14	0.2802	0.4638	0.1623	1.73×	2.86×	0.2074	0.3884	0.0711	2.92×	5.46×	0.0728	0.0754	0.0912	0.80×	0.83×
KDTree	17	16	0.2619	0.3425	0.1666	1.57×	2.06×	0.2619	0.3425	0.1666	1.57×	2.06×	–	–	–	–	–
LinearListReduction	11	11	0.0448	0.1951	0.0069	6.47×	28.16×	0.0448	0.1951	0.0069	6.47×	28.16×	–	–	–	–	–
reduceNestedList	6	5	0.1208	0.1293	0.0157	7.67×	8.21×	0.1208	0.1293	0.0157	7.67×	8.21×	–	–	–	–	–
MonoTree	3	2	0.0726	0.1518	0.0760	0.96×	2.00×	0.0214	0.0851	0.0261	0.82×	3.26×	0.0512	0.0666	0.0499	1.03×	1.34×
ObjectGraph	5	4	0.2973	0.5484	0.2786	1.07×	1.97×	0.1284	0.3491	0.0960	1.34×	3.64×	0.1688	0.1993	0.1826	0.92×	1.09×
PiecewiseFunctions	8	7	0.4994	0.6860	0.4176	1.20×	1.64×	0.1998	0.3753	0.1089	1.84×	3.45×	0.2996	0.3107	0.3088	0.97×	1.01×
TernaryTree	5	3	0.1285	0.1603	0.1208	1.06×	1.33×	0.0330	0.0551	0.0310	1.07×	1.78×	0.0955	0.1052	0.0898	1.06×	1.17×
Trie	10	9	0.2783	0.3741	0.2354	1.18×	1.59×	0.0959	0.1792	0.0378	2.53×	4.74×	0.1824	0.1949	0.1976	0.92×	0.99×
ColorOctree	25	18	0.1288	0.1611	0.0878	1.47×	1.83×	0.1288	0.1611	0.0878	1.47×	1.83×	–	–	–	–	–

Table 2: Native PAPI metrics speedup summary across all passes. For each program and metric, speedup is AoS/SoA where each side is the sum of per-pass median counter values. >1× means SoA reports fewer events.

Program	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
OctTree	1.18×	0.75×	1.12×	0.43×	0.32×	0.57×	2.53×
Compiler	1.92×	0.52×	4.35×	0.65×	17.77×	0.34×	4.60×
DBQuery	1.24×	0.47×	1.25×	0.40×	7.36×	0.24×	4.29×
DecisionTree	1.71×	0.59×	15.58×	2.25×	1.43×	2.52×	24.76×
DomTree	2.01×	0.57×	4.09×	1.26×	21.73×	0.35×	6.86×
KDTree	1.57×	0.77×	1.08×	1.02×	0.18×	1.02×	2.37×
LinearListReduction	6.46×	0.90×	39.11×	7.91×	57.64×	7.72×	9.75×
reduceNestedList	7.69×	0.53×	3.04×	0.82×	25.35×	0.82×	3.01×
MonoTree	0.98×	0.85×	0.51×	0.43×	0.31×	1.02×	1.17×
ObjectGraph	1.11×	0.75×	4.81×	0.73×	0.55×	0.80×	2.20×
PiecewiseFunctions	1.29×	0.60×	2.08×	2.14×	1.76×	0.35×	3.00×
TernaryTree	1.14×	0.79×	0.34×	0.65×	0.27×	1.02×	1.21×
Trie	1.35×	0.56×	3.35×	0.73×	2.61×	0.18×	3.85×
Geomean	1.77×	0.65×	2.63×	0.97×	2.36×	0.72×	3.66×

Table 3: Native PAPI metrics totals across all passes (baseline variants). Each entry is (Am/Sm), where Am and Sm are sums of per-pass median counter values.

Program	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
OctTree	3.2e+09/ 2.7e+09	8.2e+09 /1.1e+10	1.3e+07/ 1.1e+07	4.3e+05 /9.9e+05	8.7e+04 /2.7e+05	1.8e+04 /3.1e+04	7.0e+07/ 2.8e+07
Compiler	1.3e+09/ 7.0e+08	9.9e+08 /1.9e+09	1.3e+07/ 3.0e+06	2.2e+05 /3.5e+05	1.6e+06/ 9.2e+04	1.4e+04 /4.1e+04	5.2e+07/ 1.1e+07
DBQuery	1.1e+09/ 9.2e+08	1.1e+09 /2.2e+09	3.7e+06/ 2.9e+06	1.8e+05 /4.5e+05	7.6e+05/ 1.0e+05	9.1e+03 /3.7e+04	2.3e+07/ 5.4e+06
DecisionTree	2.6e+10/ 1.5e+10	4.0e+10 /6.8e+10	4.8e+08/ 3.1e+07	1.5e+05/ 6.8e+04	3.1e+06/ 2.2e+06	1.5e+05/ 6.0e+04	1.1e+09/ 4.5e+07
DomTree	1.2e+09/ 5.8e+08	1.2e+09 /2.2e+09	1.4e+07/ 3.5e+06	1.4e+05/ 1.1e+05	1.2e+06/ 5.5e+04	1.0e+04 /2.9e+04	5.7e+07/ 8.3e+06
KDTree	1.3e+09/ 8.1e+08	3.1e+09 /4.0e+09	1.3e+07/ 1.2e+07	1.6e+03/ 1.6e+03	2.6e+04 /1.5e+05	1.6e+03/ 1.6e+03	4.4e+07/ 1.9e+07
LinearListReduction	2.2e+08/ 3.4e+07	9.0e+07 /1.0e+08	2.0e+06/ 5.1e+04	4.5e+02/ 5.7e+01	4.1e+05/ 7.2e+03	4.5e+02/ 5.8e+01	1.2e+07/ 1.3e+06
reduceNestedList	5.9e+08/ 7.6e+07	1.2e+07 /2.3e+07	1.9e+06/ 6.3e+05	8.0e+01 /9.8e+01	1.1e+06/ 4.4e+04	8.5e+01 /1.0e+02	6.5e+06/ 2.2e+06
MonoTree	2.7e+08 /2.8e+08	9.1e+08 /1.1e+09	2.4e+05 /4.7e+05	3.3e+04 /7.5e+04	4.1e+03 /1.3e+04	2.2e+03/ 2.1e+03	2.7e+06/ 2.3e+06
ObjectGraph	1.0e+09/ 9.5e+08	3.0e+09 /4.0e+09	7.5e+06/ 1.6e+06	2.0e+05 /2.7e+05	3.0e+04 /5.4e+04	1.3e+04 /1.6e+04	2.6e+07/ 1.2e+07
PiecewiseFunctions	1.6e+09/ 1.3e+09	2.8e+09 /4.7e+09	1.4e+07/ 6.5e+06	7.7e+05/ 3.6e+05	1.8e+05/ 1.0e+05	1.4e+04 /4.1e+04	4.6e+07/ 1.5e+07
TernaryTree	4.5e+08/ 3.9e+08	1.2e+09 /1.5e+09	4.7e+05 /1.4e+05	9.0e+04 /1.4e+05	5.9e+03 /2.2e+04	3.2e+03/ 3.2e+03	3.9e+06/ 3.2e+06
Trie	8.6e+08/ 6.3e+08	1.2e+09 /2.2e+09	8.1e+06/ 2.4e+06	2.0e+05 /2.8e+05	1.4e+05/ 5.6e+04	8.9e+03 /5.0e+04	2.7e+07/ 7.0e+06

Table 4: Baseline Gibbon comparison (times only). Shown variants: AoS-mut, AoS-imm, SoA-mut. Times are full executable end-to-end wall time per run (s), not pass-sum time. **Bold** marks the fastest time across shown variants.

Program	Am	Ai	Sm
Compiler	40.80	160.1	28.24
DBQuery	33.13	41.04	29.92
DecisionTree	538.3	–	337.3
DomTree	28.62	46.73	17.05
KDTree	26.27	34.13	16.83
LinearListReduction	4.795	20.07	1.072
reduceNestedList	20.27	27.76	10.44
List	–	<i>OOM</i>	–
MonoTree	7.640	15.40	7.957
ObjectGraph	30.86	55.85	29.26
OctTree	76.26	82.99	79.27
PiecewiseFunctions	51.61	70.16	43.84
TernaryTree	13.56	16.66	12.76
Trie	29.10	38.75	25.05
ColorOctree	13.02	16.22	9.447

Table 5: Baseline Gibbon comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut. Speedups are computed from full executable end-to-end wall time per run. >1× means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm
Compiler	1.44×	3.92×	5.67×
DBQuery	1.11×	1.24×	1.37×
DecisionTree	1.60×	–	–
DomTree	1.68×	1.63×	2.74×
KDTree	1.56×	1.30×	2.03×
LinearListReduction	4.47×	4.19×	18.72×
reduceNestedList	1.94×	1.37×	2.66×
List	–	–	–
MonoTree	0.96×	2.02×	1.94×
ObjectGraph	1.05×	1.81×	1.91×
OctTree	0.96×	1.09×	1.05×
PiecewiseFunctions	1.18×	1.36×	1.60×
TernaryTree	1.06×	1.23×	1.31×
Trie	1.16×	1.33×	1.55×
ColorOctree	1.38×	1.25×	1.72×

Table 6: Per-pass performance for **OctTree**, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
barnesHutPotential	F	11/16	31%	0.0537 $\pm$ 0.00020	0.0711 $\pm$ 0.00028	0.0441$\pm$0.00012	1.22$\times$	1.32$\times$	1.61$\times$
countActive	F	10/16	38%	0.0472 $\pm$ 0.00020	0.0580 $\pm$ 0.00022	0.0308$\pm$0.00004	1.53$\times$	1.23$\times$	1.88$\times$
countParticles	F	8/16	50%	0.0475 $\pm$ 0.00027	0.0577 $\pm$ 0.00025	0.0299$\pm$0.00004	1.59$\times$	1.21$\times$	1.93$\times$
fmmPotential	F	12/16	25%	0.1089 $\pm$ 0.00074	0.1087 $\pm$ 0.00025	0.1021$\pm$0.00013	1.07 $\times$	1.00 $\times$	1.06 $\times$
sumEnergy	F	12/16	25%	0.0490 $\pm$ 0.00022	0.0673 $\pm$ 0.00023	0.0375$\pm$0.00013	1.31$\times$	1.37$\times$	1.80$\times$
sumMass	F	10/16	38%	0.0465 $\pm$ 0.00024	0.0569 $\pm$ 0.00030	0.0308$\pm$0.00050	1.51$\times$	1.22$\times$	1.85$\times$
clearFlags	M	15/16	6%	0.1834 $\pm$ 0.00045	0.1798$\pm$0.00122	0.2166 $\pm$ 0.00154	0.85 $\times$	0.98 $\times$	0.83 $\times$
scaleEnergy	M	16/16	0%	0.1807$\pm$0.00058	0.1819 $\pm$ 0.00051	0.2036 $\pm$ 0.00073	0.89 $\times$	1.01 $\times$	0.89 $\times$
paletteEntriesQuantized	F	13/25	48%	0.0623 $\pm$ 0.00034	0.0747 $\pm$ 0.00042	0.0380$\pm$0.00015	1.64$\times$	1.20$\times$	1.97$\times$
quantizationErrorProxy	F	10/25	60%	0.0665 $\pm$ 0.00038	0.0864 $\pm$ 0.00036	0.0498$\pm$0.00015	1.33$\times$	1.30$\times$	1.73$\times$
Total				0.8456	0.9424	0.7832	1.08 $\times$	1.11$\times$	1.20$\times$
Geomean						1.26$\times$	1.18$\times$	1.49$\times$	

Table 7: Per-pass PAPI counters for **OctTree** (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
barnesHutPotential	F	2.6e+08/ 2.1e+08	8.2e+08 /1.0e+09	6.2e+04 /6.2e+05	2.2e+02/ 1.7e+02	2.1e+03 /2.4e+04	2.2e+02/ 1.6e+02	6.8e+06/ 3.4e+06
countActive	F	2.3e+08/ 1.5e+08	5.8e+08/ 5.8e+08	7.8e+05/ 3.9e+05	3.2e+02/ 8.8e+01	4.6e+03 /1.3e+04	3.2e+02/ 9.2e+01	6.8e+06/ 4.9e+05
countParticles	F	2.3e+08/ 1.5e+08	5.7e+08/ 5.7e+08	7.8e+05/ 2.0e+05	3.4e+02/ 7.2e+01	2.1e+04/ 8.7e+03	3.4e+02/ 7.7e+01	6.7e+06/ 1.7e+05
fmmPotential	F	5.3e+08/ 5.0e+08	1.6e+09 /1.8e+09	2.6e+04 /7.8e+05	8.4e+02/ 7.1e+02	1.9e+03 /2.8e+04	8.3e+02/ 7.2e+02	5.3e+06/ 3.6e+06
sumEnergy	F	2.4e+08/ 1.8e+08	7.7e+08 /8.9e+08	2.6e+05 /6.8e+05	3.7e+02/ 1.4e+02	2.2e+03 /2.4e+04	3.7e+02/ 1.4e+02	7.0e+06/ 3.7e+06
sumMass	F	2.3e+08/ 1.5e+08	5.7e+08 /6.3e+08	5.9e+05/ 1.5e+05	4.8e+02/ 8.5e+01	3.2e+03 /8.4e+03	4.8e+02/ 8.6e+01	6.8e+06/ 1.6e+06
clearFlags	M	4.3e+08 /4.7e+08	8.0e+08 /1.7e+09	1.0e+06 /2.3e+06	1.8e+05 /5.7e+05	1.1e+04 /3.9e+04	7.1e+03 /1.4e+04	5.8e+06/ 3.8e+06
scaleEnergy	M	4.1e+08 /4.7e+08	9.1e+08 /1.9e+09	1.1e+06 /2.7e+06	2.4e+05 /4.2e+05	1.0e+04 /4.3e+04	7.1e+03 /1.5e+04	5.8e+06/ 4.0e+06
paletteEntriesQuantized	F	3.0e+08/ 1.8e+08	6.7e+08 /7.6e+08	4.2e+06/ 1.4e+06	5.3e+02/ 2.5e+02	1.4e+04 /4.5e+04	5.2e+02/ 2.5e+02	9.4e+06/ 2.1e+06
quantizationErrorProxy	F	3.2e+08/ 2.4e+08	9.7e+08 /1.2e+09	3.8e+06/ 2.0e+06	2.4e+02 /3.7e+02	1.7e+04 /3.6e+04	2.4e+02 /3.6e+02	9.5e+06/ 4.8e+06
Total		3.2e+09/2.7e+09	8.2e+09/1.1e+10	1.3e+07/1.1e+07	4.3e+05/9.9e+05	8.7e+04/2.7e+05	1.8e+04/3.1e+04	7.0e+07/2.8e+07

Table 8: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
blockCountPass	F	2/9	78%	0.0193 $\pm$ 0.00018	0.1181 $\pm$ 0.00041	0.0021$\pm$0.00001	9.26$\times$	6.13$\times$	56.83$\times$
branchStatsPass	F	2/9	78%	0.0189 $\pm$ 0.00018	0.1196 $\pm$ 0.00049	0.0049$\pm$0.00004	3.88$\times$	6.32$\times$	24.53$\times$
castInstCountPass	F	3/9	67%	0.0270 $\pm$ 0.00017	0.1187 $\pm$ 0.00053	0.0014$\pm$0.00001	19.26$\times$	4.40$\times$	84.69$\times$
has cycle	F	4/9	56%	0.0353 $\pm$ 0.00022	0.1409 $\pm$ 0.00071	0.0244$\pm$0.00023	1.45$\times$	4.00$\times$	5.79$\times$
instCountPass	F	2/9	78%	0.0192 $\pm$ 0.00017	0.1193 $\pm$ 0.00056	0.0022$\pm$0.00001	8.79$\times$	6.20$\times$	54.54$\times$
latencyModelPass	F	3/9	67%	0.0193 $\pm$ 0.00016	0.1188 $\pm$ 0.00048	0.0041$\pm$0.00005	4.72$\times$	6.15$\times$	29.00$\times$
memoryOpStatsPass	F	3/9	67%	0.0192 $\pm$ 0.00017	0.1187 $\pm$ 0.00046	0.0047$\pm$0.00005	4.04$\times$	6.19$\times$	25.00$\times$
throughputModelPass	F	3/9	67%	0.0192 $\pm$ 0.00017	0.1193 $\pm$ 0.00044	0.0041$\pm$0.00005	4.64$\times$	6.22$\times$	28.89$\times$
verifyIR	F	9/9	0%	0.0201 $\pm$ 0.00018	0.0998 $\pm$ 0.00039	0.0047$\pm$0.00005	4.27$\times$	4.96$\times$	21.19$\times$
stripSideEffectsPass	M	7/9	22%	0.0992$\pm$0.00035	0.2625 $\pm$ 0.00165	0.1051 $\pm$ 0.00021	0.94 $\times$	2.65$\times$	2.50$\times$
targetReturnPass	M	9/9	0%	0.1000$\pm$0.00030	0.2630 $\pm$ 0.00076	0.1093 $\pm$ 0.00023	0.91 $\times$	2.63$\times$	2.41$\times$
Total				0.3967	1.599	0.2670	1.49$\times$	4.03$\times$	5.99$\times$
Geomean						3.87$\times$	4.85$\times$	18.73$\times$	

Table 9: Per-pass PAPI counters for **Compiler** (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
blockCountPass	F	9.4e+07/ 1.0e+07	5.1e+07/ 4.5e+07	1.4e+06/ 5.7e+03	3.2e+02/ 4.1e+01	3.0e+05/ 1.1e+03	3.2e+02/ 4.2e+01	4.8e+06/ 2.6e+04
branchStatsPass	F	9.3e+07/ 2.3e+07	7.9e+07 /8.6e+07	8.6e+05/ 2.2e+04	3.1e+02/ 5.2e+01	7.7e+04/ 3.1e+03	3.0e+02/ 5.4e+01	4.8e+06/ 7.0e+05
castInstCountPass	F	1.3e+08/ 6.8e+06	5.1e+07/ 3.9e+07	1.4e+04 /2.2e+04	1.0e+02/ 4.2e+01	8.8e+02 /1.2e+03	1.0e+02/ 4.3e+01	4.8e+06/ 1.6e+04
has cycle	F	1.7e+08/ 1.2e+08	1.1e+08 /1.5e+08	2.4e+06/ 2.9e+05	5.5e+02/ 1.7e+02	1.8e+05/ 2.3e+04	5.1e+02/ 1.7e+02	5.5e+06/ 2.7e+06
instCountPass	F	9.4e+07/ 1.0e+07	5.6e+07 /5.7e+07	1.4e+06/ 1.1e+04	2.9e+02/ 5.0e+01	2.9e+05/ 1.2e+03	2.9e+02/ 5.1e+01	4.8e+06/ 2.9e+04
latencyModelPass	F	9.4e+07/ 2.0e+07	6.2e+07 /6.9e+07	1.3e+06/ 3.6e+04	3.2e+02/ 4.5e+01	2.7e+05/ 3.9e+03	3.2e+02/ 4.6e+01	4.8e+06/ 7.3e+05
memoryOpStatsPass	F	9.4e+07/ 2.3e+07	8.5e+07 /9.1e+07	6.6e+05/ 2.6e+04	3.1e+02/ 5.0e+01	4.7e+04/ 3.1e+03	3.1e+02/ 5.1e+01	4.8e+06/ 6.9e+05
throughputModelPass	F	9.3e+07/ 2.0e+07	6.2e+07 /6.9e+07	1.3e+06/ 3.3e+04	4.6e+02/ 5.2e+01	2.7e+05/ 3.5e+03	4.3e+02/ 5.3e+01	4.8e+06/ 7.3e+05
verifyIR	F	9.8e+07/ 2.3e+07	7.4e+07 /8.6e+07	1.0e+06/ 1.6e+04	3.2e+02/ 1.0e+02	1.8e+05/ 4.0e+03	3.2e+02/ 1.0e+02	4.8e+06/ 7.0e+05
stripSideEffectsPass	M	1.9e+08 /2.1e+08	1.8e+08 /6.0e+08	1.5e+06/ 1.1e+06	1.2e+05 /1.4e+05	7.1e+03 /2.2e+04	7.0e+03 /1.6e+04	4.0e+06/ 2.2e+06
targetReturnPass	M	1.9e+08 /2.3e+08	1.7e+08 /6.3e+08	1.3e+06 /1.4e+06	1.0e+05 /2.0e+05	6.9e+03 /2.6e+04	4.4e+03 /2.4e+04	4.0e+06/ 2.7e+06
Total		1.3e+09/7.0e+08	9.9e+08/1.9e+09	1.3e+07/3.0e+06	2.2e+05/3.5e+05	1.6e+06/9.2e+04	1.4e+04/4.1e+04	5.2e+07/1.1e+07

Table 10: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
countJoins	F	3/15	80%	0.0246 $\pm$ 0.00013	0.0359 $\pm$ 0.00014	0.0118$\pm$0.00002	2.09$\times$	1.46$\times$	3.06$\times$
filterSelectivitySkew	F	5/15	67%	0.0250 $\pm$ 0.00030	0.0354 $\pm$ 0.00018	0.0136$\pm$0.00003	1.84$\times$	1.41$\times$	2.61$\times$
hashJoinPressure	F	5/15	67%	0.0254 $\pm$ 0.00013	0.0398 $\pm$ 0.00013	0.0135$\pm$0.00002	1.87$\times$	1.57$\times$	2.94$\times$
sumCost	F	6/15	60%	0.0245 $\pm$ 0.00013	0.0355 $\pm$ 0.00051	0.0156$\pm$0.00004	1.57$\times$	1.45$\times$	2.27$\times$
sumMemory	F	6/15	60%	0.0245 $\pm$ 0.00030	0.0357 $\pm$ 0.00014	0.0158$\pm$0.00004	1.55$\times$	1.46$\times$	2.26$\times$
sumRows	F	6/15	60%	0.0329 $\pm$ 0.00011	0.0372 $\pm$ 0.00015	0.0233$\pm$0.00003	1.41$\times$	1.13$\times$	1.60$\times$
clearQueryFlags	M	14/15	7%	0.0823$\pm$0.00030	0.0895 $\pm$ 0.00022	0.0952 $\pm$ 0.00015	0.86 $\times$	1.09 $\times$	0.94 $\times$
scaleCosts	M	15/15	0%	0.0827$\pm$0.00029	0.0902 $\pm$ 0.00061	0.0978 $\pm$ 0.00019	0.85 $\times$	1.09 $\times$	0.92 $\times$
Total				0.3218	0.3993	0.2865	1.12$\times$	1.24$\times$	1.39$\times$
Geomean						1.44$\times$	1.32$\times$	1.90$\times$	

Table 11: Per-pass PAPI counters for DBQuery (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
countJoins	F	1.2e+08/ 5.7e+07	7.4e+07/ 7.1e+07	6.5e+05/ 2.5e+04	6.8e+02/ 1.3e+02	1.6e+05/ 2.0e+03	6.8e+02/ 9.7e+01	3.0e+06/ 7.3e+04
filterSelectivitySkew	F	1.2e+08/ 6.6e+07	1.0e+08 /1.1e+08	4.7e+05/ 7.2e+04	2.1e+02/ 1.0e+02	1.1e+05/ 6.2e+03	2.1e+02/ 1.0e+02	3.0e+06/ 4.3e+05
hashJoinPressure	F	1.2e+08/ 6.6e+07	8.4e+07 /9.9e+07	6.4e+05/ 8.9e+04	2.6e+02/ 1.0e+02	1.6e+05/ 7.3e+03	2.6e+02/ 1.0e+02	3.0e+06/ 2.2e+05
sumCost	F	1.2e+08/ 7.6e+07	9.0e+07 /1.2e+08	6.1e+05/ 8.0e+04	3.8e+02/ 8.5e+01	1.4e+05/ 1.2e+04	3.9e+02/ 8.6e+01	3.0e+06/ 5.5e+05
sumMemory	F	1.2e+08/ 7.7e+07	9.0e+07 /1.2e+08	5.2e+05/ 8.3e+04	3.5e+02/ 1.1e+02	1.1e+05/ 1.2e+04	3.6e+02/ 1.1e+02	3.0e+06/ 5.4e+05
sumRows	F	1.6e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.3e+05/ 1.1e+05	1.9e+02/ 1.3e+02	2.1e+04/ 7.6e+03	1.9e+02/ 1.3e+02	2.8e+06/ 4.8e+05
clearQueryFlags	M	1.9e+08 /2.3e+08	1.9e+08 /7.2e+08	2.6e+05 /1.2e+06	9.4e+04 /2.5e+05	2.8e+04/ 2.6e+04	3.5e+03 /1.4e+04	2.6e+06/ 1.4e+06
scaleCosts	M	1.9e+08 /2.4e+08	2.2e+08 /7.5e+08	2.7e+05 /1.2e+06	8.3e+04 /1.9e+05	3.0e+04 /3.0e+04	3.5e+03 /2.2e+04	2.7e+06/ 1.7e+06
Total		1.1e+09/9.2e+08	1.1e+09/2.2e+09	3.7e+06/2.9e+06	1.8e+05/4.5e+05	7.6e+05/1.0e+05	9.1e+03/3.7e+04	2.3e+07/5.4e+06

Table 12: Per-pass performance for DecisionTree, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
classify Batch	F	5/7	29%	4.649 $\pm$ 0.0031	–	2.810$\pm$0.0245	1.65$\times$	–	–
classify Depth	F	4/7	43%	0.0669 $\pm$ 0.00037	–	0.0391$\pm$0.00006	1.71$\times$	–	–
classify tree	F	5/7	29%	0.0667 $\pm$ 0.00044	–	0.0393$\pm$0.00006	1.70$\times$	–	–
countFeatureUses	F	3/7	57%	0.0729 $\pm$ 0.00040	–	0.0387$\pm$0.00013	1.88$\times$	–	–
countLeaves	F	2/7	71%	0.0688 $\pm$ 0.00030	–	0.0271$\pm$0.00009	2.54$\times$	–	–
countNodes	F	2/7	71%	0.0686 $\pm$ 0.00033	–	0.0285$\pm$0.00005	2.41$\times$	–	–
countSmallLeaves	F	3/7	57%	0.0766 $\pm$ 0.00050	–	0.0385$\pm$0.00016	1.99$\times$	–	–
inferenceCost	F	2/7	71%	0.0716 $\pm$ 0.00032	–	0.0310$\pm$0.00004	2.31$\times$	–	–
sumImpurity	F	3/7	57%	0.0729 $\pm$ 0.00069	–	0.0384$\pm$0.00014	1.90$\times$	–	–
sumPathLengths	F	3/7	57%	0.0763 $\pm$ 0.00035	–	0.0407$\pm$0.00013	1.88$\times$	–	–
sumSamples	F	3/7	57%	0.0714 $\pm$ 0.00054	–	0.0380$\pm$0.00012	1.88$\times$	–	–
treeDepth	F	2/7	71%	0.0721 $\pm$ 0.00038	–	0.0304$\pm$0.00005	2.37$\times$	–	–
Total				5.434	–	3.200	1.70$\times$	–	–
Geomean						2.00$\times$	–	–	

Table 13: Per-pass PAPI counters for DecisionTree (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
classify Batch	F	2.3e+10/ 1.4e+10	3.3e+10 /5.9e+10	4.1e+08/ 2.6e+07	1.3e+05/ 5.8e+04	2.7e+06/ 2.0e+06	1.3e+05/ 5.0e+04	9.6e+08/ 3.0e+07
classify Depth	F	3.3e+08/ 1.9e+08	4.7e+08 /8.5e+08	5.9e+06/ 3.6e+05	1.7e+03/ 5.5e+02	3.4e+04/ 2.8e+04	1.7e+03/ 5.4e+02	1.4e+07/ 4.4e+05
classify tree	F	3.3e+08/ 1.9e+08	4.7e+08 /8.5e+08	5.9e+06/ 3.6e+05	1.8e+03/ 7.1e+02	3.1e+04/ 2.9e+04	1.8e+03/ 6.5e+02	1.4e+07/ 4.4e+05
countFeatureUses	F	3.6e+08/ 1.9e+08	7.5e+08 /7.8e+08	6.3e+06/ 1.0e+06	1.9e+03/ 1.4e+03	3.1e+04/ 1.4e+04	1.8e+03/ 1.4e+03	1.5e+07/ 2.4e+06
countLeaves	F	3.3e+08/ 1.3e+08	6.0e+08 /6.1e+08	6.7e+06/ 8.8e+04	1.9e+03/ 9.1e+02	6.1e+04/ 7.8e+03	1.9e+03/ 8.4e+02	1.4e+07/ 5.0e+05
countNodes	F	3.3e+08/ 1.4e+08	6.1e+08 /6.2e+08	6.6e+06/ 9.4e+04	1.3e+03/ 5.1e+02	6.2e+04/ 7.9e+03	1.3e+03/ 4.7e+02	1.4e+07/ 5.1e+05
countSmallLeaves	F	3.7e+08/ 1.9e+08	7.1e+08 /8.4e+08	5.9e+06/ 2.0e+05	1.8e+03/ 1.2e+03	3.3e+04/ 1.3e+04	1.8e+03/ 1.1e+03	1.5e+07/ 2.4e+06
inferenceCost	F	3.5e+08/ 1.5e+08	6.1e+08/ 6.1e+08	7.0e+06/ 1.1e+05	2.4e+03/ 5.7e+02	2.1e+04/ 7.8e+03	2.3e+03/ 5.6e+02	1.4e+07/ 5.0e+05
sumImpurity	F	3.5e+08/ 1.9e+08	7.6e+08 /7.8e+08	6.5e+06/ 1.0e+06	1.8e+03/ 1.1e+03	3.4e+04/ 1.3e+04	1.8e+03/ 1.1e+03	1.5e+07/ 2.4e+06
sumPathLengths	F	3.7e+08/ 2.0e+08	8.0e+08 /9.0e+08	6.8e+06/ 1.1e+06	2.3e+03/ 1.4e+03	4.1e+04/ 1.4e+04	2.2e+03/ 1.4e+03	1.5e+07/ 2.4e+06
sumSamples	F	3.5e+08/ 1.9e+08	6.4e+08 /7.7e+08	5.7e+06/ 4.5e+05	2.2e+03/ 1.1e+03	4.0e+04/ 1.4e+04	2.2e+03/ 1.1e+03	1.5e+07/ 2.4e+06
treeDepth	F	3.5e+08/ 1.5e+08	6.1e+08/ 6.1e+08	7.0e+06/ 1.0e+05	2.2e+03/ 6.0e+02	2.3e+04/ 8.1e+03	2.2e+03/ 5.9e+02	1.4e+07/ 5.0e+05
Total		2.6e+10/1.5e+10	4.0e+10/6.8e+10	4.8e+08/3.1e+07	1.5e+05/6.8e+04	3.1e+06/2.2e+06	1.5e+05/6.0e+04	1.1e+09/4.5e+07

Table 14: Per-pass performance for DomTree, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
count styled	F	3/15	80%	0.0517 $\pm$ 0.00027	0.0977 $\pm$ 0.00042	0.0131$\pm$0.00006	3.95$\times$	1.89$\times$	7.45$\times$
find max Bottom	F	5/15	67%	0.0523 $\pm$ 0.00033	0.0963 $\pm$ 0.00054	0.0219$\pm$0.00008	2.38$\times$	1.84$\times$	4.39$\times$
SumArea	F	6/15	60%	0.0524 $\pm$ 0.00030	0.0965 $\pm$ 0.00042	0.0226$\pm$0.00010	2.32$\times$	1.84$\times$	4.27$\times$
sumTextWidth	F	3/15	80%	0.0510 $\pm$ 0.00034	0.0979 $\pm$ 0.00044	0.0134$\pm$0.00005	3.80$\times$	1.92$\times$	7.29$\times$
computeWidths	M	14/15	7%	0.0370$\pm$0.00016	0.0380 $\pm$ 0.00013	0.0508 $\pm$ 0.00012	0.73 $\times$	1.03 $\times$	0.75 $\times$
scaleLayout	M	15/15	0%	0.0358$\pm$0.00013	0.0374 $\pm$ 0.00012	0.0404 $\pm$ 0.00010	0.89 $\times$	1.05 $\times$	0.93 $\times$
Total				0.2802	0.4638	0.1623	1.73$\times$	1.66$\times$	2.86$\times$
Geomean						1.94$\times$	1.54$\times$	2.98$\times$	

Table 15: Per-pass PAPI counters for DomTree (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
count styled	F	2.5e+08/ 6.4e+07	2.6e+08 /2.6e+08	2.8e+06/ 3.4e+04	1.6e+03/ 1.2e+02	3.3e+05/ 5.0e+03	1.6e+03/ 1.2e+02	1.4e+07/ 8.5e+05
find max Bottom	F	2.6e+08/ 1.1e+08	2.5e+08 /4.1e+08	3.2e+06/ 6.7e+05	2.3e+03/ 1.0e+02	1.7e+05/ 9.8e+03	2.3e+03/ 1.0e+02	1.4e+07/ 2.2e+06
SumArea	F	2.6e+08/ 1.1e+08	2.9e+08 /4.9e+08	3.1e+06/ 5.5e+05	8.6e+02/ 7.5e+01	1.8e+05/ 1.3e+04	8.6e+02/ 7.6e+01	1.4e+07/ 2.9e+06
sumTextWidth	F	2.5e+08/ 6.5e+07	2.0e+08 /2.4e+08	3.0e+06/ 2.1e+04	1.6e+03/ 1.2e+02	4.8e+05/ 4.3e+03	1.6e+03/ 1.2e+02	1.4e+07/ 8.8e+05
computeWidths	M	8.0e+07 /1.4e+08	1.4e+08 /4.6e+08	1.1e+06 /1.5e+06	5.8e+04/ 4.5e+04	5.5e+03 /1.3e+04	1.9e+03 /1.5e+04	1.3e+06/ 6.7e+05
scaleLayout	M	7.6e+07 /9.6e+07	8.8e+07 /2.9e+08	9.8e+05/ 7.3e+05	7.1e+04/ 6.2e+04	1.5e+04/ 9.5e+03	1.8e+03 /1.3e+04	1.3e+06/ 8.1e+05
Total		1.2e+09/5.8e+08	1.2e+09/2.2e+09	1.4e+07/3.5e+06	1.4e+05/1.1e+05	1.2e+06/5.5e+04	1.0e+04/2.9e+04	5.7e+07/8.3e+06

Table 16: Per-pass performance for KDTree, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
countInRange tight_box	F	13/17	24%	0.0413 $\pm$ 0.00026	0.0564 $\pm$ 0.00031	0.0248$\pm$0.00016	1.66$\times$	1.36$\times$	2.27$\times$
Find nearest Neighbour	F	13/17	24%	0.0416 $\pm$ 0.00027	0.0563 $\pm$ 0.00027	0.0262$\pm$0.00013	1.59$\times$	1.35$\times$	2.15$\times$
photonMappingBenchmark	F	12/17	29%	0.0574 $\pm$ 0.00029	0.0643 $\pm$ 0.00028	0.0453$\pm$0.00010	1.27$\times$	1.12$\times$	1.42$\times$
pointCloudNeighborhood	F	11/17	35%	0.0379 $\pm$ 0.00024	0.0545 $\pm$ 0.00024	0.0206$\pm$0.00017	1.84$\times$	1.44$\times$	2.65$\times$
sumMassInRange	F	14/17	18%	0.0409 $\pm$ 0.00025	0.0561 $\pm$ 0.00027	0.0256$\pm$0.00016	1.59$\times$	1.37$\times$	2.19$\times$
twoPointCorrelation bin_8-16	F	11/17	35%	0.0428 $\pm$ 0.00027	0.0549 $\pm$ 0.00027	0.0241$\pm$0.00009	1.77$\times$	1.28$\times$	2.28$\times$
Total				0.2619	0.3425	0.1666	1.57$\times$	1.31$\times$	2.06$\times$
Geomean						1.61$\times$	1.32$\times$	2.12$\times$	

Table 17: Per-pass PAPI counters for `KDTree` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
countInRange tight_box	F	2.0e+08/ 1.2e+08	4.0e+08 /5.5e+08	1.9e+06 /2.2e+06	2.6e+02 /2.6e+02	3.3e+03 /2.7e+04	2.6e+02 /2.6e+02	7.5e+06/ 3.4e+06
Find nearest Neighbour	F	2.0e+08/ 1.3e+08	4.7e+08 /6.3e+08	2.5e+06/ 2.0e+06	1.5e+02 /1.6e+02	4.4e+03 /2.5e+04	1.5e+02 /1.6e+02	7.6e+06/ 3.3e+06
photonMappingBenchmark	F	2.8e+08/ 2.2e+08	1.0e+09 /1.2e+09	2.0e+06 /2.5e+06	5.4e+02 /6.4e+02	4.4e+03 /2.3e+04	5.4e+02 /6.4e+02	6.2e+06/ 2.7e+06
pointCloudNeighborhood	F	1.8e+08/ 1.0e+08	3.3e+08 /4.8e+08	2.7e+06/ 1.4e+06	2.5e+02/ 1.7e+02	5.8e+03 /2.0e+04	2.5e+02/ 1.7e+02	7.7e+06/ 2.9e+06
sumMassInRange	F	2.0e+08/ 1.3e+08	4.0e+08 /5.7e+08	2.3e+06 /3.5e+06	1.3e+02 /2.1e+02	4.3e+03 /3.5e+04	1.3e+02 /2.1e+02	7.6e+06/ 3.6e+06
twoPointCorrelation bin_8_16	F	2.1e+08/ 1.2e+08	4.8e+08 /6.1e+08	1.9e+06/ 7.3e+05	2.8e+02/ 1.3e+02	3.5e+03 /1.6e+04	2.8e+02/ 1.3e+02	7.4e+06/ 2.7e+06
Total		1.3e+09/8.1e+08	3.1e+09/4.0e+09	1.3e+07/1.2e+07	1.6e+03/1.6e+03	2.6e+04/1.5e+05	1.6e+03/1.6e+03	4.4e+07/1.9e+07

Table 18: Per-pass performance for `LinearListReduction`, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
reduction	F	2/11	82%	0.0448 $\pm$ 0.00031	0.1951 $\pm$ 0.00061	0.0069$\pm$0.00005	6.47$\times$	4.35$\times$	28.16$\times$
Total				0.0448	0.1951	0.0069	6.47$\times$	4.35$\times$	28.16$\times$
Geomean						6.47$\times$	4.35$\times$	28.16$\times$	

Table 19: Per-pass PAPI counters for `LinearListReduction` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
reduction	F	2.2e+08/ 3.4e+07	9.0e+07 /1.0e+08	2.0e+06/ 5.1e+04	4.5e+02/ 5.7e+01	4.1e+05/ 7.2e+03	4.5e+02/ 5.8e+01	1.2e+07/ 1.3e+06
Total		2.2e+08/3.4e+07	9.0e+07/1.0e+08	2.0e+06/5.1e+04	4.5e+02/5.7e+01	4.1e+05/7.2e+03	4.5e+02/5.8e+01	1.2e+07/1.3e+06

Table 20: Per-pass performance for `reduceNestedList`, ADT has 6 fields, SoA uses 5 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
reduction	F	2/6	67%	0.1208 $\pm$ 0.00051	0.1293 $\pm$ 0.00049	0.0157$\pm$0.00014	7.67$\times$	1.07 $\times$	8.21$\times$
Total				0.1208	0.1293	0.0157	7.67$\times$	1.07 $\times$	8.21$\times$
Geomean						7.67$\times$	1.07 $\times$	8.21$\times$	

Table 21: Per-pass PAPI counters for `reduceNestedList` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
reduction	F	5.9e+08/ 7.6e+07	1.2e+07 /2.3e+07	1.9e+06/ 6.3e+05	8.0e+01 /9.8e+01	1.1e+06/ 4.4e+04	8.5e+01 /1.0e+02	6.5e+06/ 2.2e+06
Total		5.9e+08/7.6e+07	1.2e+07/2.3e+07	1.9e+06/6.3e+05	8.0e+01/9.8e+01	1.1e+06/4.4e+04	8.5e+01/1.0e+02	6.5e+06/2.2e+06

Table 22: Per-pass performance for `MonoTree`, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
sumTree	F	3/3	0%	0.0109$\pm$0.00006	0.0424 $\pm$ 0.00006	0.0132 $\pm$ 0.00005	0.83 $\times$	3.88$\times$	3.21$\times$
sumTree TailRec	F	3/3	0%	0.0105$\pm$0.00006	0.0427 $\pm$ 0.00005	0.0129 $\pm$ 0.00006	0.81 $\times$	4.08$\times$	3.31$\times$
add1Tree	M	3/3	0%	0.0512 $\pm$ 0.00023	0.0666 $\pm$ 0.00014	0.0499$\pm$0.00012	1.03 $\times$	1.30$\times$	1.34$\times$
Total				0.0726	0.1518	0.0760	0.96 $\times$	2.09$\times$	2.00$\times$
Geomean						0.88 $\times$	2.74$\times$	2.42$\times$	

Table 23: Per-pass PAPI counters for `MonoTree` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
sumTree	F	5.3e+07 /6.4e+07	2.2e+08 /2.6e+08	1.3e+04 /3.1e+04	2.8e+02 /3.7e+02	8.1e+02 /3.9e+03	2.8e+02 /3.6e+02	1.0e+06/ 8.4e+05
sumTree TailRec	F	5.1e+07 /6.2e+07	1.9e+08 /2.2e+08	8.2e+03 /1.8e+04	3.0e+02 /3.8e+02	8.4e+02 /3.9e+03	3.0e+02 /3.8e+02	1.0e+06/ 8.5e+05
add1Tree	M	1.7e+08/ 1.5e+08	5.0e+08 /5.9e+08	2.2e+05 /4.2e+05	3.2e+04 /7.5e+04	2.4e+03 /5.5e+03	1.6e+03/ 1.4e+03	7.1e+05/ 6.4e+05
Total		2.7e+08/2.8e+08	9.1e+08/1.1e+09	2.4e+05/4.7e+05	3.3e+04/7.5e+04	4.1e+03/1.3e+04	2.2e+03/2.1e+03	2.7e+06/2.3e+06

Table 24: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
countLargeItems	F	3/5	40%	0.0182 $\pm$ 0.00014	0.0508 $\pm$ 0.00018	0.0128$\pm$0.00006	1.42$\times$	2.80$\times$	3.97$\times$
countMarked	F	3/5	40%	0.0182 $\pm$ 0.00012	0.0506 $\pm$ 0.00013	0.0127$\pm$0.00004	1.44$\times$	2.77$\times$	3.99$\times$
countSurvivors	F	4/5	20%	0.0192 $\pm$ 0.00015	0.0500 $\pm$ 0.00013	0.0158$\pm$0.00008	1.22$\times$	2.60$\times$	3.17$\times$
deadBytes	F	4/5	20%	0.0189 $\pm$ 0.00009	0.0500 $\pm$ 0.00012	0.0152$\pm$0.00007	1.25$\times$	2.65$\times$	3.30$\times$
liveBytes	F	4/5	20%	0.0191 $\pm$ 0.00010	0.0486 $\pm$ 0.00012	0.0151$\pm$0.00007	1.27$\times$	2.54$\times$	3.22$\times$
sumObjIds	F	3/5	40%	0.0176 $\pm$ 0.00014	0.0495 $\pm$ 0.00011	0.0123$\pm$0.00006	1.43$\times$	2.81$\times$	4.02$\times$
totalHeapSize	F	3/5	40%	0.0172 $\pm$ 0.00016	0.0497 $\pm$ 0.00016	0.0122$\pm$0.00026	1.42$\times$	2.89$\times$	4.09$\times$
sweepUnmarked	M	5/5	0%	0.0851$\pm$0.00023	0.1005 $\pm$ 0.00024	0.0917 $\pm$ 0.00034	0.93 $\times$	1.18$\times$	1.10 $\times$
touchHotObjects	M	5/5	0%	0.0838$\pm$0.00077	0.0988 $\pm$ 0.00017	0.0909 $\pm$ 0.00013	0.92 $\times$	1.18$\times$	1.09 $\times$
Total				0.2973	0.5484	0.2786	1.07 $\times$	1.84$\times$	1.97$\times$
Geomean						1.24$\times$	2.26$\times$	2.80$\times$	

Table 25: Per-pass PAPI counters for `ObjectGraph` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
countLargeItems	F	8.8e+07/ 6.2e+07	2.6e+08 /2.8e+08	8.8e+05/ 5.1e+04	3.7e+02/ 2.4e+02	3.2e+03 /3.4e+03	3.8e+02/ 2.4e+02	3.1e+06/ 8.7e+05
countMarked	F	8.9e+07/ 6.2e+07	2.6e+08 /2.8e+08	9.0e+05/ 5.2e+04	3.7e+02/ 2.1e+02	3.3e+03 /3.8e+03	3.8e+02/ 2.1e+02	3.1e+06/ 8.8e+05
countSurvivors	F	9.3e+07/ 7.7e+07	3.1e+08 /3.8e+08	7.4e+05/ 8.7e+04	7.2e+02/ 3.4e+02	2.7e+03 /5.3e+03	7.2e+02/ 3.4e+02	3.0e+06/ 1.6e+06
deadBytes	F	9.2e+07/ 7.4e+07	2.7e+08 /3.5e+08	5.7e+05/ 6.8e+04	3.8e+02/ 2.4e+02	2.3e+03 /5.3e+03	3.8e+02/ 2.4e+02	3.0e+06/ 1.7e+06
liveBytes	F	9.3e+07/ 7.4e+07	2.7e+08 /3.5e+08	5.1e+05/ 6.7e+04	4.4e+02/ 3.0e+02	2.0e+03 /5.3e+03	4.4e+02/ 3.0e+02	3.0e+06/ 1.6e+06
sumObjIds	F	8.6e+07/ 6.0e+07	2.5e+08 /2.7e+08	1.0e+06/ 7.9e+04	7.6e+02/ 2.1e+02	3.2e+03 /4.3e+03	7.6e+02/ 2.1e+02	3.1e+06/ 8.5e+05
totalHeapSize	F	8.4e+07/ 5.8e+07	2.5e+08 /2.7e+08	9.8e+05/ 9.8e+04	3.4e+02/ 2.0e+02	3.2e+03 /3.5e+03	3.4e+02/ 1.9e+02	3.1e+06/ 8.6e+05
sweepUnmarked	M	2.1e+08 /2.4e+08	5.5e+08 /8.8e+08	9.9e+05/ 5.1e+05	8.6e+04 /1.4e+05	4.6e+03 /1.2e+04	4.6e+03 /7.3e+03	2.3e+06/ 1.7e+06
touchHotObjects	M	2.1e+08 /2.4e+08	5.7e+08 /9.0e+08	9.5e+05/ 5.5e+05	1.1e+05 /1.3e+05	5.1e+03 /1.2e+04	4.5e+03 /6.5e+03	2.2e+06/ 1.7e+06
Total		1.0e+09/9.5e+08	3.0e+09/4.0e+09	7.5e+06/1.6e+06	2.0e+05/2.7e+05	3.0e+04/5.4e+04	1.3e+04/1.6e+04	2.6e+07/1.2e+07

Table 26: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
autorefineMaxLevel	F	4/8	50%	0.0303 $\pm$ 0.00016	0.0580 $\pm$ 0.00027	0.0234$\pm$0.00004	1.30$\times$	1.91$\times$	2.48$\times$
compressMass	F	3/8	62%	0.0295 $\pm$ 0.00026	0.0593 $\pm$ 0.00042	0.0133$\pm$0.00004	2.21$\times$	2.01$\times$	4.44$\times$
lbDeuxLoadProxy	F	5/8	38%	0.0321 $\pm$ 0.00025	0.0609 $\pm$ 0.00025	0.0221$\pm$0.00008	1.46$\times$	1.90$\times$	2.76$\times$
norm2Estimate	F	5/8	38%	0.0357 $\pm$ 0.00025	0.0689 $\pm$ 0.00020	0.0215$\pm$0.00008	1.66$\times$	1.93$\times$	3.21$\times$
pmapCutHistogram	F	4/8	50%	0.0424 $\pm$ 0.00018	0.0686 $\pm$ 0.00021	0.0154$\pm$0.00008	2.76$\times$	1.62$\times$	4.47$\times$
truncateTolViolations	F	3/8	62%	0.0298 $\pm$ 0.00024	0.0595 $\pm$ 0.00027	0.0133$\pm$0.00006	2.25$\times$	1.99$\times$	4.49$\times$
addConstPW	M	8/8	0%	0.1504$\pm$0.00055	0.1545 $\pm$ 0.00077	0.1553 $\pm$ 0.00031	0.97 $\times$	1.03 $\times$	0.99 $\times$
diffPW	M	8/8	0%	0.1491$\pm$0.00047	0.1563 $\pm$ 0.00038	0.1535 $\pm$ 0.00027	0.97 $\times$	1.05 $\times$	1.02 $\times$
Total				0.4994	0.6860	0.4176	1.20$\times$	1.37$\times$	1.64$\times$
Geomean						1.59$\times$	1.63$\times$	2.58$\times$	

Table 27: Per-pass PAPI counters for `PiecewiseFunctions` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
autorefineMaxLevel	F	1.5e+08/ 1.1e+08	2.5e+08 /3.7e+08	2.3e+06/ 3.1e+05	8.3e+02/ 9.3e+01	7.4e+03/ 6.8e+03	8.4e+02/ 9.7e+01	6.2e+06/ 1.4e+06
compressMass	F	1.4e+08/ 6.5e+07	2.5e+08 /2.8e+08	2.1e+06/ 2.7e+04	1.1e+03/ 5.0e+02	1.8e+04/ 5.2e+03	1.1e+03/ 5.0e+02	6.3e+06/ 8.5e+05
lbDeuxLoadProxy	F	1.6e+08/ 1.1e+08	3.3e+08 /4.9e+08	2.2e+06/ 5.4e+05	7.2e+02/ 2.0e+02	1.4e+04/ 1.0e+04	7.2e+02/ 2.1e+02	6.3e+06/ 2.2e+06
norm2Estimate	F	1.7e+08/ 1.0e+08	3.4e+08 /4.4e+08	1.1e+06/ 6.6e+05	4.7e+02/ 3.3e+02	1.0e+04/ 9.9e+03	4.7e+02/ 3.3e+02	6.1e+06/ 2.2e+06
pmapCutHistogram	F	2.1e+08/ 7.5e+07	2.8e+08 /3.7e+08	1.4e+06/ 6.6e+04	1.8e+03/ 7.1e+02	8.5e+04/ 7.4e+03	1.7e+03/ 7.1e+02	5.3e+06/ 1.6e+06
truncateTolViolations	F	1.4e+08/ 6.4e+07	2.5e+08 /2.9e+08	2.2e+06/ 4.0e+04	1.4e+03/ 4.0e+02	1.6e+04/ 4.9e+03	1.4e+03/ 4.0e+02	6.3e+06/ 8.5e+05
addConstPW	M	3.3e+08 /3.7e+08	5.7e+08 /1.2e+09	1.1e+06 /2.5e+06	4.0e+05/ 1.9e+05	1.3e+04 /2.9e+04	3.4e+03 /2.1e+04	4.8e+06/ 3.1e+06
diffPW	M	3.3e+08 /3.7e+08	5.9e+08 /1.3e+09	1.2e+06 /2.4e+06	3.6e+05/ 1.7e+05	1.3e+04 /2.7e+04	4.4e+03 /1.7e+04	4.8e+06/ 3.1e+06
Total		1.6e+09/1.3e+09	2.8e+09/4.7e+09	1.4e+07/6.5e+06	7.7e+05/3.6e+05	1.8e+05/1.0e+05	1.4e+04/4.1e+04	4.6e+07/1.5e+07

Table 28: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
sum tree	F	5/5	0%	0.0330 $\pm$ 0.00009	0.0551 $\pm$ 0.00009	0.0310$\pm$0.00011	1.07 $\times$	1.67$\times$	1.78$\times$
add 1 tree	M	5/5	0%	0.0955 $\pm$ 0.00036	0.1052 $\pm$ 0.00028	0.0898$\pm$0.00024	1.06 $\times$	1.10$\times$	1.17$\times$
Total				0.1285	0.1603	0.1208	1.06 $\times$	1.25$\times$	1.33$\times$
Geomean						1.06 $\times$	1.36$\times$	1.44$\times$	

Table 29: Per-pass PAPI counters for `TernaryTree` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
sum tree	F	1.6e+08/ 1.5e+08	4.6e+08 /5.4e+08	8.0e+04 /3.3e+05	9.6e+01 /6.2e+02	9.5e+02 /1.0e+04	1.0e+02 /6.2e+02	2.1e+06/ 1.8e+06
add 1 tree	M	2.9e+08/ 2.4e+08	7.5e+08 /9.8e+08	3.9e+05 /1.1e+06	9.0e+04 /1.4e+05	5.0e+03 /1.2e+04	3.1e+03/ 2.5e+03	1.8e+06/ 1.5e+06
Total		4.5e+08/3.9e+08	1.2e+09/1.5e+09	4.7e+05/1.4e+06	9.0e+04/1.4e+05	5.9e+03/2.2e+04	3.2e+03/3.2e+03	3.9e+06/3.2e+06

Table 30: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
autocompleteTopKProxy	F	5/10	50%	0.0209 $\pm$ 0.00014	0.0363 $\pm$ 0.00019	0.0100$\pm$0.00007	2.09$\times$	1.73$\times$	3.62$\times$
countLazyNodes	F	4/10	60%	0.0197 $\pm$ 0.00014	0.0371 $\pm$ 0.00016	0.0080$\pm$0.00003	2.47$\times$	1.89$\times$	4.65$\times$
countTerminals	F	3/10	70%	0.0184 $\pm$ 0.00008	0.0351 $\pm$ 0.00015	0.0067$\pm$0.00003	2.74$\times$	1.91$\times$	5.23$\times$
sumPrefixFreq	F	3/10	70%	0.0184 $\pm$ 0.00016	0.0354 $\pm$ 0.00036	0.0065$\pm$0.00005	2.83$\times$	1.93$\times$	5.45$\times$
sumSubtreeHints	F	3/10	70%	0.0185 $\pm$ 0.00012	0.0353 $\pm$ 0.00020	0.0066$\pm$0.00044	2.79$\times$	1.90$\times$	5.32$\times$
decayTrieStats	M	10/10	0%	0.0915$\pm$0.00032	0.0969 $\pm$ 0.00028	0.1005 $\pm$ 0.00028	0.91 $\times$	1.06 $\times$	0.96 $\times$
resetTraversalState	M	10/10	0%	0.0909$\pm$0.00023	0.0981 $\pm$ 0.00037	0.0971 $\pm$ 0.00016	0.94 $\times$	1.08 $\times$	1.01 $\times$
Total				0.2783	0.3741	0.2354	1.18$\times$	1.34$\times$	1.59$\times$
Geomean						1.92$\times$	1.59$\times$	3.06$\times$	

Table 31: Per-pass PAPI counters for `Trie` (Am/Sm). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Am/Sm)	INS (Am/Sm)	L1D (Am/Sm)	L1I (Am/Sm)	L2D (Am/Sm)	L2I (Am/Sm)	LLC (Am/Sm)
autocompleteTopKProxy	F	1.0e+08/ 4.8e+07	1.3e+08 /2.1e+08	1.3e+06/ 3.9e+05	1.0e+02/ 6.5e+01	3.6e+04/ 5.0e+03	1.1e+02/ 6.5e+01	4.0e+06/ 1.2e+06
countLazyNodes	F	9.6e+07/ 3.9e+07	1.3e+08 /1.6e+08	1.2e+06/ 2.4e+04	1.4e+02/ 1.0e+02	1.4e+04/ 3.4e+03	1.5e+02/ 1.0e+02	4.2e+06/ 8.4e+05
countTerminals	F	8.9e+07/ 3.3e+07	1.0e+08 /1.2e+08	1.3e+06/ 1.0e+04	2.0e+02/ 8.1e+01	2.9e+04/ 2.3e+03	2.1e+02/ 8.0e+01	4.1e+06/ 4.4e+05
sumPrefixFreq	F	8.9e+07/ 3.1e+07	1.1e+08 /1.2e+08	1.3e+06/ 1.3e+04	1.7e+02/ 7.2e+01	2.3e+04/ 2.5e+03	1.8e+02/ 7.2e+01	4.1e+06/ 4.3e+05
sumSubtreeHints	F	9.0e+07/ 3.2e+07	1.1e+08 /1.2e+08	1.3e+06/ 1.3e+04	2.0e+02/ 7.5e+01	2.8e+04/ 2.5e+03	2.1e+02/ 7.6e+01	4.1e+06/ 4.3e+05
decayTrieStats	M	1.9e+08 /2.3e+08	3.5e+08 /7.9e+08	8.2e+05 /1.2e+06	8.6e+04 /1.3e+05	6.5e+03 /2.4e+04	3.6e+03 /3.1e+04	3.3e+06/ 2.1e+06
resetTraversalState	M	2.0e+08 /2.2e+08	2.8e+08 /7.0e+08	8.4e+05/ 7.2e+05	1.2e+05 /1.5e+05	7.8e+03 /1.6e+04	4.5e+03 /1.9e+04	3.2e+06/ 1.6e+06
Total		8.6e+08/6.3e+08	1.2e+09/2.2e+09	8.1e+06/2.4e+06	2.0e+05/2.8e+05	1.4e+05/5.6e+04	8.9e+03/5.0e+04	2.7e+07/7.0e+06